

Mohammad Raffi

Embedded Cybersecurity Engineer

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PROFESSIONAL SUMMARY

- Embedded Cybersecurity Engineer with hands-on experience in securing ARM Cortex-M based embedded systems across industrial automation and IoT domains in both India and the USA.
- Strong expertise in designing and implementing secure boot architectures to establish root-of-trust and prevent execution of unauthorized firmware.
- Proven experience implementing firmware signing, authentication, and integrity validation using industry-standard cryptographic algorithms such as AES, RSA, ECC, and SHA.
- Extensive hands-on work with secure firmware update mechanisms (FOTA/OTA), including encryption, version control, rollback protection, and power-failure recovery handling.
- Solid experience in implementing TLS-based secure communication over TCP/IP to protect device-to-cloud and device-to-gateway data exchanges.
- Application security-focused engineer with experience securing backend services and APIs interacting with devices and cloud platforms.
- Exposure to securing applications deployed on Microsoft Azure, including identity management, secure configuration, and cloud access controls.
- Understanding of application security principles for web and API-based applications, including authentication, authorization, and secure data handling.
- Experienced in developing real-time embedded firmware using Embedded C/C++ on FreeRTOS and bare-metal environments while maintaining security and performance balance.
- Strong collaboration experience with hardware, validation, and manufacturing teams to ensure secure startup, reset behavior, and device provisioning.
- Experienced in applying Secure SDLC practices across design, development, testing, and deployment phases.

EDUCATION

- *Master's in computer information technology - Elmhurst University, Illinois - May 2025*
- *Bachelors in Electronics and Communication Engineering - Narasaraopeta Engineering College - April 2023*

PROFESSIONAL EXPERIENCE

Role: Embedded Cybersecurity Engineer

Client: Molex, Lisle, IL

Aug 2024 – Present

Project Description

Worked on securing embedded firmware for industrial electronic modules and intelligent interconnect devices used in automation and smart manufacturing systems. The project focused on implementing end-to-end embedded security including secure boot, encrypted firmware updates, secure communication, and device identity protection while maintaining real-time performance and reliability.

Responsibilities

- Designed and implemented a secure boot architecture on ARM Cortex-M microcontrollers by establishing a root-of-trust and validating firmware integrity during power-up to prevent execution of unauthorized or tampered code.
- Implemented firmware image signing and verification using ECC/RSA cryptographic algorithms, ensuring only authenticated firmware images are accepted during boot and update processes.

- Developed a secure OTA (FOTA) firmware update framework with encryption, version validation, and rollback protection to safely support field updates without risking device bricking.
- Took ownership of application-layer security for backend services and APIs interacting with embedded devices and cloud platforms.
- Reviewed application and API security controls to ensure proper authentication, authorization, encryption, and secure data flow.
- Integrated TLS-based secure communication over TCP/IP to protect data exchanged between embedded devices, gateways, and backend services from interception and man-in-the-middle attacks.
- Implemented AES-based encryption for sensitive configuration data, credentials, and runtime parameters stored in internal and external flash memory.
- Supported application security in Microsoft Azure environments by working with identity management, access controls, and secure handling of secrets.
- Performed threat modeling (TARA) to identify attack surfaces related to bootloader, memory access, debug ports, communication interfaces, and update mechanisms.
- Hardened embedded firmware by disabling unused peripherals, removing debug access in production builds, and permanently locking JTAG/SWD interfaces.
- Played an active role in security incident response by analyzing issues, identifying root causes, coordinating mitigation, and documenting corrective and preventive actions.
- Assisted in evaluating and rolling out security tools, working closely with vendors and internal engineering teams.
- Designed and implemented secure key management strategies, including protected storage of cryptographic keys and certificates using hardware-supported memory protection features.
- Collaborated with hardware and validation teams to verify secure startup behavior, fault handling, and reset recovery scenarios under abnormal operating conditions.
- Conducted secure code reviews and static analysis to detect buffer overflows, unsafe pointer usage, race conditions, and improper memory access.
- Supported penetration testing and vulnerability assessments by reproducing reported issues, analyzing root causes, and implementing firmware-level mitigations.
- Implemented unique device identity provisioning during manufacturing to support authentication, secure onboarding, and lifecycle tracking of devices.
- Optimized cryptographic operations to meet real-time constraints and low-power requirements without impacting system responsiveness.
- Ensured firmware design aligned with industrial cybersecurity best practices, drawing from IEC 62443 concepts and secure SDLC principles.
- Helped develop security guidelines, standard operating procedures, and secure development practices for engineering teams.
- Created and maintained security architecture documentation, threat models, and mitigation plans for audits, customer reviews, and long-term maintenance.

Role: Embedded Firmware & Security Engineer

Client: Efftronics Systems Pvt. Ltd., India

Jun 2021 – Jul 2023

Project Description

Worked on embedded firmware development for industrial automation and IoT-based monitoring systems. The role gradually evolved from core firmware development into implementing foundational embedded security features such as encrypted communication, secure firmware updates, and interface hardening.

Responsibilities

- Developed embedded firmware in C for ARM-based microcontrollers used in industrial control and monitoring devices.

- Designed bootloader and application separation architecture to support firmware upgrades, recovery mechanisms, and field servicing.
- Implemented firmware integrity checks during boot and update cycles to detect corrupted or unauthorized images.
- Integrated TLS-based encrypted communication to secure device-to-server data transfer over Ethernet and IP networks.
- Implemented MQTT communication with authentication, enabling reliable and secure telemetry exchange for IoT-based systems.
- Worked with cross-functional teams to improve security practices, documentation, and secure development workflows.
- Applied AES encryption to protect stored credentials, device configuration data, and calibration parameters in non-volatile memory.
- Assisted in implementing firmware update mechanisms over UART and network interfaces with basic validation and error handling.
- Performed interface-level hardening for UART, SPI, and I2C communications to reduce exposure to unauthorized access.
- Involved in security issue analysis by investigating reported problems, identifying root causes, implementing fixes, and documenting preventive actions.
- Conducted firmware validation testing, including power-loss, reset recovery, and long-run stability testing.
- Identified and resolved memory leaks, stack overflows, timing issues, and race conditions in real-time applications.
- Followed Secure SDLC practices by integrating security considerations during design, development, testing, and deployment activities.
- Prepared technical documentation covering firmware architecture, update procedures, and security controls.
- Built strong fundamentals in embedded cybersecurity, forming the foundation for advanced secure firmware work in later roles.
- Applied application security principles while designing software components, ensuring secure authentication, authorization, and data handling.